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Broadband mid-infrared optical limiter based on vanadium dioxide phase transition

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Optical limiters (OLs) are necessary devices for the protection of sensitive photo-detectors from high-intensity laser irradiation. However, current OLs based on vanadium dioxide (VO_2) phase transition show a limited wavelength protection bandwidth, insufficient transmittance in the low-temperature phase, and an open-limited turndown ratio (infrared switching rate). Therefore, this study presented a continuously tunable 3000–5000 nm mid-infrared broadband VO_2-based OL by combining the insulator-to-metal transition of VO_2 with a multi-layer thin film. And the optical-thermal properties of the OL were analyzed using the finite element method (FEM). The results revealed that the OL represented excellent laser protection performance in the 3000–5000 nm broadband wavelength region such as a low-temperature phase transmittance higher than 70%, an infrared switching rate of 98%, and the broad working bandwidth of more than 3000 nm. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION

In recent decades, there has been growing attention to the threat of high-power laser radiation to photosensitive systems. Optical limiting is a technique to protect photosensitive devices from damage caused by intense optical radiation or laser blinding attacks [1–3]. Optical limiters (OLs) that can mitigate the damage to delicate optical components under high-intensity illumination are therefore designed to block high-intensity laser radiation (limiting state) while transmitting low-intensity light (open state) [4,5]. Most optical limiters use materials with nonlinear absorption, which are transparent to low-intensity light but turn opaque if the light intensity exceeds a certain threshold level. A photonic limiter usually involves a nonlinear or a phase-change material (PCM). At low intensities, the photonic limiter displays strong transmittance. High-intensity radiation, however, forces nonlinearity to kick in or phase transition to be induced. Over the past few decades, many OLs based on nonlinear optical (NLO) materials have been proposed. Although NLO materials are inexpensive, they are not ideal because they actuate slowly and exhibit an unfavorable balance between the open-state transmittance and turndown ratio [6–8].

Vanadium dioxide (VO_2), as a leading insulator-to-metal phase transition (IMT) material, is considered a promising alternative to laser protection material due to the characteristics of thermal-induced and photo-induced phase transition [9–11]. Vanadium dioxide also demonstrates the pronounced infrared switching behavior across the phase transition [12–19]. Therefore, VO_2-based OLs have attracted considerable interest for their overwhelming performance. Currently, VO_2-based optical limiters have been widely studied. For example, Kats *et al.* [20] demonstrated an optical limiter for the mid-infrared region with a tunable limiting threshold based on an electrically driving metal meta-surface. Guan *et al.* [21] combined the VO_2 insulator-to-metal transition with double-sided HfO_2 anti-reflection layers to achieve high open-state transmittance and a large switching ratio. Howes *et al.* recently demonstrated an optical limiter by incorporating VO_2 with dielectric meta-surfaces, which had a low triggering threshold [22]. However, practical applications still face how to enable a VO_2-based optical limiter to coordinate both high and low-temperature phase transmittance and a large turndown ratio (infrared switching rate) between the VO_2 phase transition, while preserving a fast phase transition response time and a broad protection bandwidth in laser protection. These have become an important bottleneck, restricting the application of vanadium dioxide thin films in the mid-infrared field of laser protection.

This paper presented a broadband mid-infrared optical limiter based on vanadium dioxide multi-layer thin films. The optical-thermal transient response properties of the optical limiter were rigorously analyzed using the finite element method. The dependence of the vanadium dioxide's complex refractive index on temperature was examined, and the transmission characteristics of the optical limiter across various wavelengths as a function of temperature were derived. Opto-thermal simulations provided insights into the limiter performance such as the limiting threshold and the response time. The findings demonstrated that the optical limiter exhibited an extended protection bandwidth and enhanced infrared switching performance.

Table 1. Parameter Values Used to Fit the Complex Dielectric Function of VO_2 [29]

	E_{Pj}, E_{Oj} (eV)						τ_j (eV^{-1})						
Energy state	ε_0	E_{P1}	E_{o1}	τ_1	E_{p2}	E_{o2}	τ_2	E_{p3}	E_{o3}	τ_3	E_{p4}	E_{o4}	τ_4
Metallic phase	4.4	3.6	0	1.3	3.2	2.8	1.2	3.6	3.5	1.2	3.8	4.5	1
Semiconductor phase	5.5	1.65	1.25	1.1	2.8	2.8	1.2	4.25	3.6	0.9	5.6	4.9	0.5

2. DESIGN AND SIMULATION

A. Optical Constants of Vanadium Dioxide in the Mid-Infrared Region

The optical constants of VO_2 directly affect light transmission and reflection characteristics, which in turn determine the overall performance of the optical limiter. Therefore, it is crucial to accurately determine the optical constants of vanadium dioxide in the 3000–5000 nm wavelength range. The Drude–Lorentz dispersion model [23,24] allows for the description of the complex refractive index ε of VO_2 in its semiconductor and metallic phases, as described by Eqs. (1) and (2). When photon energy E exceeds 1.2 eV, VO_2 is in a semiconductor phase, which can be analyzed using Eq. (1). For photon energy E less than 1.2 eV, VO_2 corresponds to its metallic phase, as shown by Eq. (2):

Drude-Lorentz dispersion model

$$\varepsilon = \varepsilon_0 - \frac{E_{P1}^2}{E^2 + \frac{iE}{\tau_j}} - \sum_{j=1}^{4} \frac{E_{pj}^2}{E^2 - E_{Oj}^2 + \frac{iE}{\tau_j}}, \quad (1)$$

$$\varepsilon = \varepsilon_0 - \sum_{j=1}^{4} \frac{E_{pj}^2}{E^2 - E_{Oj}^2 + \frac{iE}{\tau_j}}, \quad (2)$$

where ε_0 represents the constant contribution of the dielectric function for high-frequency electron transitions, $\frac{E_{P1}^2}{E^2+\frac{iE}{\tau_j}}$ follows the standard of the Drude model and depends on the material's free carriers, E_{Pj} denotes the plasma energy, τ_j represents the collision time, $\sum_{j=1}^{4} \frac{E_{pj}^2}{E^2-E_{Oj}^2+\frac{iE}{\tau_j}}$ depends on the inter-band transitions, and E_{oj} represents the resonant energy. The parameter values used to fit the dielectric function of VO_2 are listed in Table 1.

The dielectric function ε, refractive index n, and extinction coefficient k of vanadium dioxide are related by Eq. (3):

$$\mathrm{Re}(\varepsilon) - i\mathrm{Im}(\varepsilon) = (n - ik)^2. \quad (3)$$

The connection between photon energy E (eV) and wavelength λ (nm) is described by Eq. (4):

$$E = hc/\lambda = 1242/\lambda. \quad (4)$$

According to formulas (1), (2), (3), and (4), the variation of the refractive index (n) and the extinction coefficient (k) of vanadium dioxide (VO_2) in its semiconductor phase (low temperature) and metallic phase (high temperature) with respect to the incident light wavelength can be determined.

Figure 1 illustrates the refractive index and the extinction coefficient of VO_2 in the low- and high-temperature states, respectively. From Fig. 1, we can see that VO_2 maintained consistent optical performance within the 3000–5000 nm wavelength range in the low-temperature state (semiconductor phase). And the refractive index was approximately 3.17, and the extinction coefficient was around 0.01. In contrast, when vanadium dioxide was in the high-temperature state, corresponding to its metallic phase, both the refractive index and the extinction coefficient showed a remarkable increase in the 3000–5000 nm wavelength range. This suggests a reflectance and absorption increase of VO_2, leading to decreased transmission characteristics, thereby achieving efficient optical limiting.

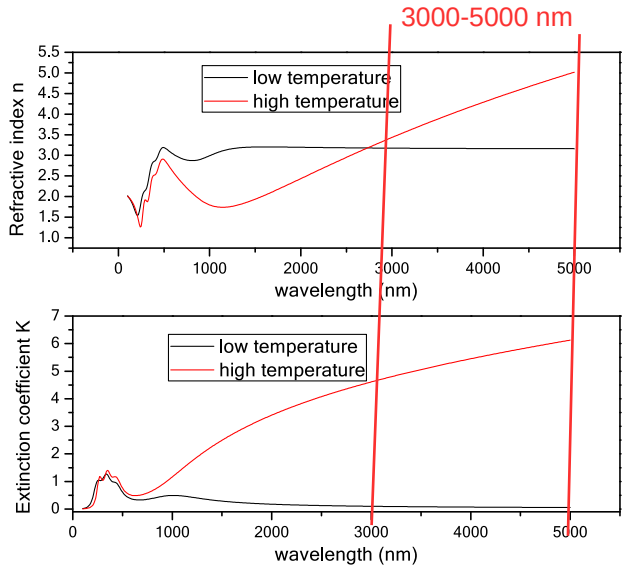

Fig. 1. Refractive index(n) and extinction coefficient (k) of VO_2 in the low- and high-temperature states.

B. Structural Models

Based on the relationship between the optical constants of VO_2 and wavelength mentioned above, the optical limiter working in the 3000–5000 nm wavelength range, consisting of a multi-layer thin film, was designed using film design software. First, sapphire (Al_2O_3) was selected as the substrate for the optical limiter. Then, we employed continuous optimization techniques alongside target-specific optimization of the film stack through variable materials and the equivalent film stack calculation method. And a needle optimization procedure was employed to iteratively modify thin-film thickness and material configurations for progressive approximation to desired spectral properties. The selected materials in the film stack included germanium (Ge), zinc selenide (ZnSe), yttrium fluoride (YF_3), hafnium oxide (HfO_2), and silicon dioxide (SiO_2). First, in our design, attributed to the selected materials that are almost transparent in the mid- infrared region, absorption losses in adjacent layers are very low, and the anti-reflection layer was composed of materials with alternating high (H) and low (L) refractive indices, which also made interface reflection almost

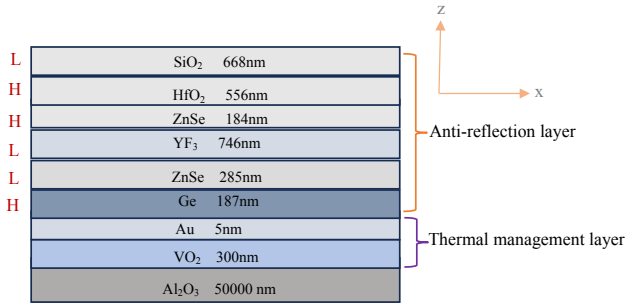

Fig. 2. Structure model of the optical limiter consisting of a composite multilayer film.

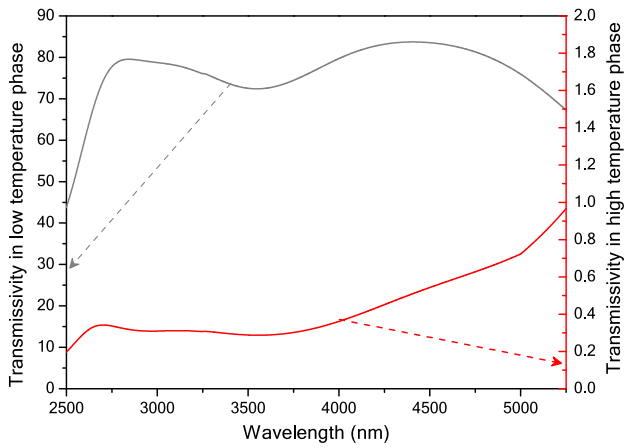

Fig. 3. Transmittance of the device at the high- and low-temperature phases of VO_2.

negligible. Second, to assure the compatibility of the materials in the film fabrication process, an SiO_2 layer was used as the outermost layer of the film stack in film design, mainly to avoid the occurrence of high stress when YF_3 was placed on the outermost layer. At the same time, to match SiO_2, hafnium dioxide was used instead of zinc selenide in the sub-outer layer. Finally, the thin gold layer (Au) acted as a heat source to enhance the phase transition of vanadium dioxide, thus providing a foundation for the subsequent analysis of the optical-thermal coupling within the film system. The optimized design model of the broadband optical limiter is illustrated in Fig. 2.

Figure 3 shows the transmittance of the optimized optical limiter at both open and limiting states. The result revealed that at the open state, the optical limiter maintained a transmittance higher than 70% across the wavelength range of 2665–5177 nm, with a maximum transmittance value of 83.7% at a wavelength of 4405 nm. High open-state transmittance ensures that the optical limiter had high transmission efficiency for low-intensity lasers. In contrast, at the limiting state, the transmittance of the optical limiter was only about 1%. Low limiting-state transmittance indicates that the optical limiter can effectively limit the high-intensity laser irradiation.

Additionally, the open-limited turndown ratio (infrared switching rate) K can be derived according to formula (5):

$$K = \frac{(T_L - T_H)}{T_L}, \tag{5}$$

where T_L and T_H represent the transmittance of the optical limiter under low-temperature and high-temperature phases, respectively. It is revealed that the open-limited turndown ratio (infrared switching rate) K of the optical limiter ran to 98% with a bandwidth of 4405 nm. Thus, the optical limiter demonstrated the pronounced infrared switching behavior across the entire working bandwidth.

C. Optical-Thermal Simulations

To comprehensively analyze the optical-thermal characteristics of the optical limiter, optical-thermal response under dynamic excitation conditions was conducted. And a fully coupled optical-thermal model was developed to accurately simulate the response of the optical limiter under dynamic illumination by applying COMSOL Multiphysics.

Our model consisted of two-dimensional geometry to reduce computation time. The laser beam used in our model had a Gaussian intensity distribution, and incident light of different wavelengths was irradiated perpendicularly onto the structure. Due to the transverse symmetry of the structure along the x-direction, the periodic boundary conditions on the y-direction were introduced in the optical problem simulation. The optical problem was solved with a frequency-domain study, and the solution allowed the computation of the absorbed power density. The heat generation within the device was due to the absorbed laser power density in the Au and VO_2 layers. Additionally, the thermal problem was solved using a transient solution method by solving the thermal diffusion equations [25,26]. A convection boundary condition $q \cdot m = h(T - T_{ext})$ was accounted on the interface of air–SiO_2, where q is the heat flux, m is the unit normal vector pointing outside the domain, $h \approx 25\ \mathrm{W/m^2K}$ is the convective heat transfer coefficient, $T_{ext} = 293$ K is the external ambient temperature. On the lateral sides, we assign adiabatic boundary conditions. In the modeling, we neglected the variations of thermal conductivity (K_T) and mass density (ρ) of the vanadium dioxide across the phase transition. The simulation parameters are reported in Table 2, where ρ represents the density, K_T is the thermal conductivity, and C_P denotes the specific heat capacity at constant pressure.

The obtained temperature was used to adjust the dielectric function of the VO_2 layer. When the VO_2 layer contains both semiconductor and metallic regions, the dielectric function of

Table 2. Simulation Parameters of Materials

Parameter	Al_2O_3	VO_2	Au	SiO_2	HfO_2	ZnSe	YF_3	Ge
ρ (kg/m^3)	3950	4340	19300	2200	5990	5270	4010	5320
K_T (W/m · k)	25	6	317	1.4	1.32	18	3.2	0.599
C_P (J/kg · k)	775	696	129	695	1960	399	920	23.2

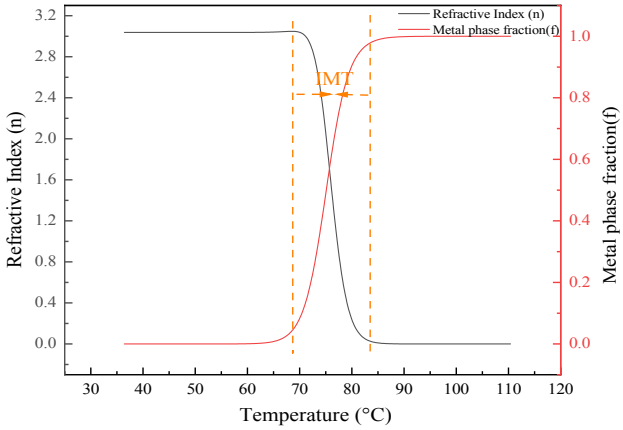

Fig. 4. Metallic phase fraction (f), refractive index (n) of vanadium dioxide under different temperatures.

the vanadium dioxide film can be expressed using the Looyenga mixing rule based on effective medium theory (EMT) [27,28] as follows:

$$[\varepsilon(T)]^S = [1 - f(T)]\varepsilon_{\text{sem}}^s + f(T)\varepsilon_{\text{met}}^s, \tag{6}$$

$$f(T) = \left\{1 + \exp\left[\frac{W}{K_B}\left(\frac{1}{T} - \frac{1}{T_C}\right)\right]\right\}^{-1}, \tag{7}$$

where ε_{sem} and ε_{met} are the complex dielectric functions of vanadium dioxide in the semiconductor and metallic phase, respectively; f represents the temperature-dependent metallic fraction of VO_2; and $s = 1/3$, $W = 4.85$ eV, $Tc = 75.1°C$.

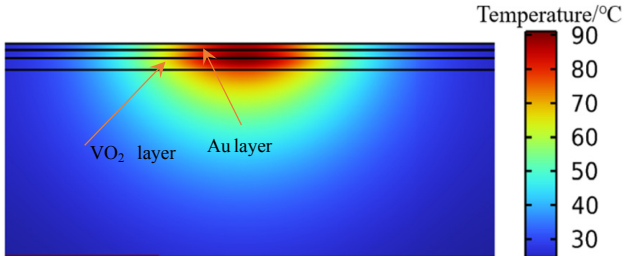

Fig. 5. Temperature profile inside the multilayer films.

According to formulas (6) and (7), the complex refractive index of vanadium dioxide was derived, which is closely related to both temperature and the metallic phase fraction. As shown in Fig. 4, when vanadium dioxide transitioned from a semiconductor phase to a metallic phase, the refractive index of vanadium dioxide decreased, and the volume fraction of the vanadium dioxide metallic phase reached 1 with the increasing temperature.

3. RESULTS AND DISCUSSIONS

Opto-thermal simulations provide insights into limiter performance, which can be used to estimate the limiting threshold and the response time of the device. The first investigation involved applying a single pulse with varying intensity to the film system to determine the laser protection capability of optical limiter in the 3000–5000 nm wavelength range. Figure 5 shows the temperature distribution diagram under the incident laser pulse with the intensity of 60 MW/cm^2 at a wavelength of 3000 nm.

It can be seen that the presence of the gold layer caused most of the heat to be concentrated at the Au-VO_2 interface after the laser pulse arrived, resulting in vanadium dioxide arriving at a higher temperature rapidly. Consequently, the gold layer served as an effective heat source for the vanadium dioxide, and over time, the heat gradually diffused to the substrate, and the temperature along the vertical direction decreased gradually. Therefore, the gold film provided an efficient way to increase the radiation absorption inside the VO_2 and induced thermal switching on shorter timescales.

Figure 6 illustrates the time-resolved temperature and transmittance of the limiter with different laser pulse intensities at a wavelength of 3000 nm and a pulse width of 60 ps. When the pulse intensity was below 50 MW/cm^2, the vanadium dioxide did not reach the critical temperature of the phase transition; thus, the optical limiter remained in the "open" state. For laser intensity exceeding 50 MW/cm^2, the optical limiter arrived to the "limiting" state, and the maximum temperature of the

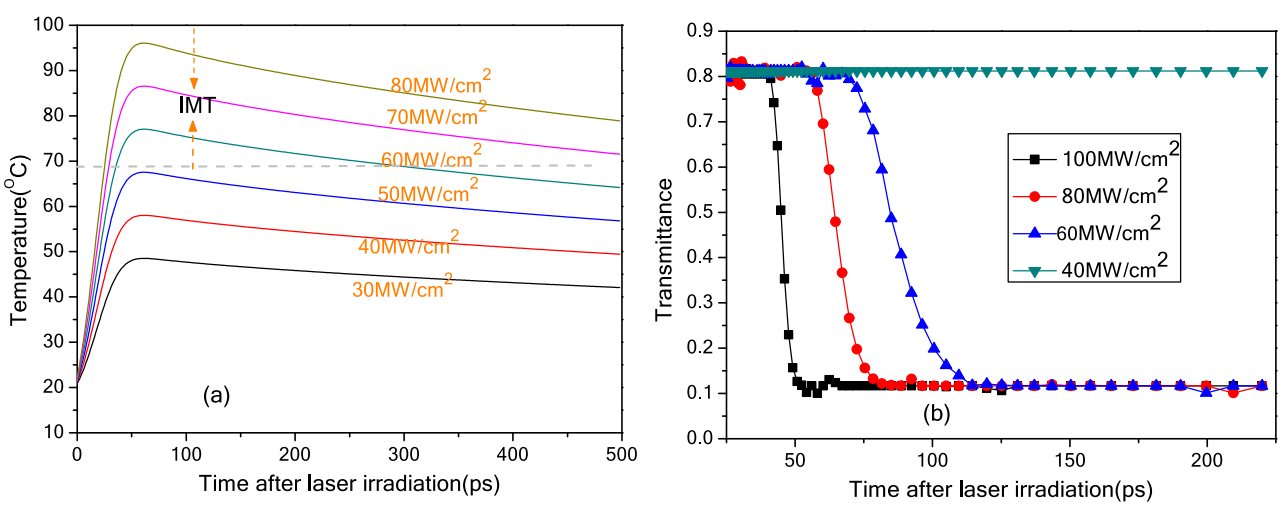

Fig. 6. Simulated time-resolved temperature and transmittance of the limiter for different incident intensities at fixed pulse-width of 60 ps.

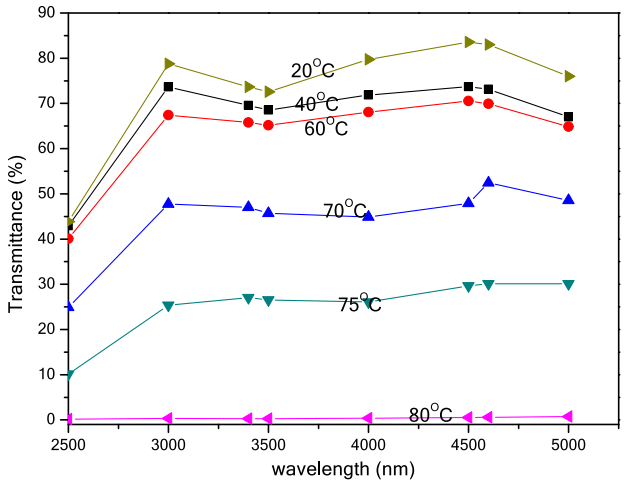

Fig. 7. Transmittance as a function of wavelength at different temperatures.

vanadium dioxide layer hit the peak after the pulse arrived approximately 50 ps later. Given the incident laser intensity of 60 MW/cm^2, the response time (from the open state to the limiting state) was about 75 ps, which can be obtained from Fig. 6(b); at this time, the maximum temperature exceeded the critical value (68°C) of VO_2 phase transition. The response time reduced to 50 ps with the incident intensity greater than 80 MW/cm^2. Therefore, when exposed to high-intensity laser irradiation, the optical limiter can respond quickly and effectively provide protection and prevent damage to the sensitive optoelectronic detectors.

Based on the changes in the complex refractive index of vanadium dioxide (VO_2) at different temperatures (Fig. 4) and their impact on optical transmittance, the relationship between transmittance and wavelength at various temperatures was also derived. Figure 7 shows the transmittance of the optical limiter with wavelengths at different temperatures. When the temperature is below the phase transition temperature of vanadium dioxide, the variation in the complex dielectric function is minimal, resulting in higher transmittance of the optical limiter. However, when the temperature reaches or exceeds the phase transition temperature, there is a significant change in the complex dielectric function of vanadium dioxide, leading to a substantial decrease in transmittance.

Figure 8 shows the temperature and transmittance variation of the limiter irradiated by laser pulse with the intensity of 100 MW/cm^2 under different laser wavelengths. Analysis indicates that the limiting state of the optical limiter will be triggered with higher incident intensity, and the response time maybe much longer as the laser wavelength increases.

In Fig. 9, we reported the temperature as a function of time for different pulse durations at the same laser energy density. For pulse durations smaller than 5 ps, the maximum temperature is achieved about 30 ps after laser irradiation. After the laser irradiation stopped, the temperature followed an exponential decay due to heat diffusion. While for the pulse duration longer than 5 ps, the maximum temperature was below the critical temperature of the VO_2 phase transition. So, the IMT in VO_2 can be triggered on timescales as short as a few picoseconds using high-intensity laser pulses.

4. CONCLUSION

This study presented a broadband mid-infrared optical limiter within the 3000–5000 nm wavelength range based on the phase transition properties of vanadium dioxide (VO_2). To validate its performance, the optical limiter underwent optical-thermal analysis using the finite element method. The results demonstrated that the optical limiter maintained a higher transmittance than 70% at the open state across the wavelength range of 3000–5000 nm, with a peak transmittance value of 83.7% at a wavelength of 4405 nm. Additionally, the optical limiter exhibited an infrared switching rate of 98% with a broad bandwidth of higher than 3000 nm. These studies indicate that the optical limiter features high open-state transmittance, low limiting-state transmittance, broad working bandwidth, and exceptional infrared switching rate, significantly enhancing its application potential in the 3000–5000 nm mid-infrared laser protection region.

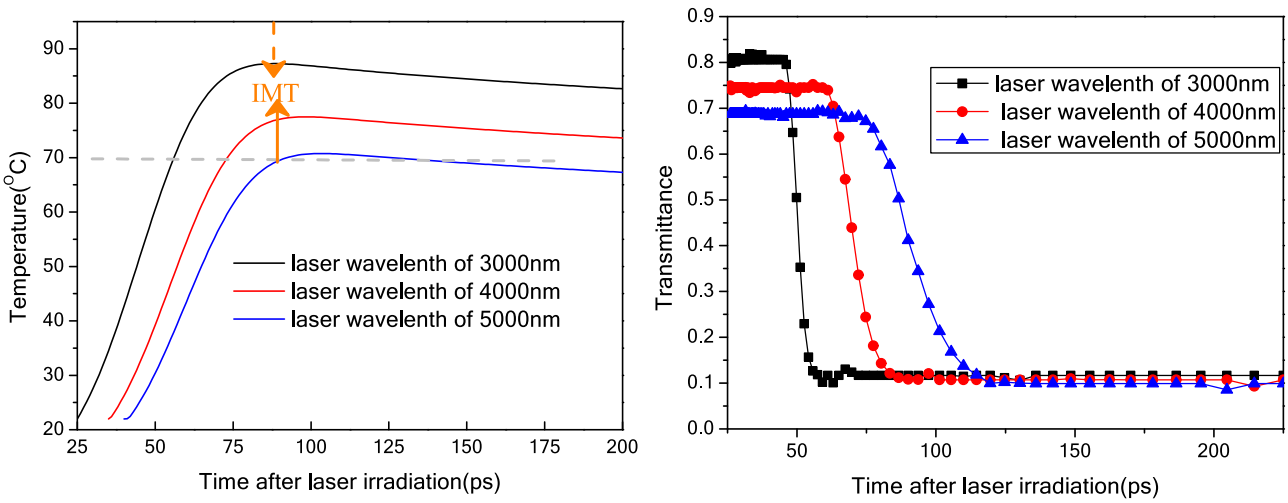

Fig. 8. Time-resolved temperature and transmittance irradiated by an incident intensity of 100 MW/cm^2 for different wavelengths.

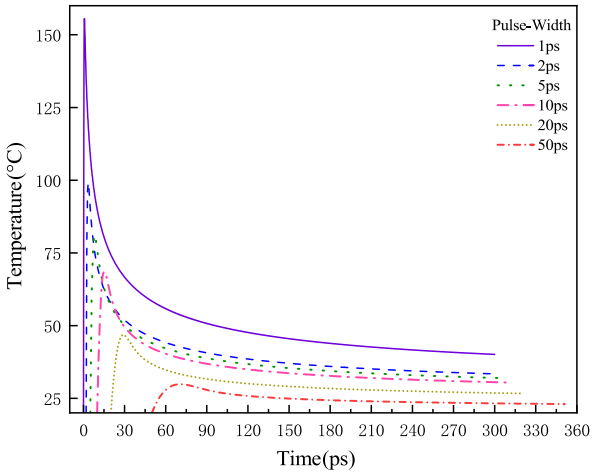

Fig. 9. Time-resolved temperature for different pulse durations at laser energy density of 100 J/m^2.

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Disclosures. The authors declare no conflicts of interest.

Data availability. The data that support the findings of this study are available within the article.

REFERENCES

1. W. Jiang, T. Zheng, B. Wu, *et al.*, "A versatile photodetector assisted by photovoltaic and bolometric effects," Light Sci. Appl. **9**, 160 (2020).
2. S. Chandra, D. Franklin, J. Cozart, *et al.*, "Adaptive multispectral infrared camouflage," ACS Photon. **5**, 4513–4519 (2018).
3. W.-Y. Liang, F. Liu, Y.-J. Lu, *et al.*, "High optical nonlinearity in low-dimensional halide perovskite polycrystalline films," Opt. Express **28**, 24919–24927 (2020).
4. J. Huang, Y. Lu, L. Zhou, *et al.*, "Design of photonic crystal nonlinear laser power limiter based on topological edge states and optical Kerr effect," Opt. Express **28**, 12080–12092 (2020).
5. J. Parra, J. Navarro-Arenas, M. Menghini, *et al.*, "Low-threshold power and tunable integrated optical limiter based on an ultracompact VO_2/Si waveguide," APL Photonics **6**, 121301 (2021).
6. Y. A. Mezenov, N. K. Kulachenkov, A. N. Yankin, *et al.*, "Polymer matrix incorporated with ZIF-8 for application in nonlinear optics," Nanomaterials **10**, 1036 (2020).
7. Y. Dai, Y. Wang, S. Das, *et al.*, "Electrical control of interband resonant nonlinear optics in monolayer MoS_2," ACS Nano **14**, 8442 (2020).
8. K. Usui, K. Matsumoto, E. Katayama, *et al.*, "A deformable low-threshold optical limiter with oligothiophene-doped liquid crystals," ACS Appl. Mater. Interfaces **13**, 23049–23056 (2021).
9. Y. Guo, B. Xiong, Y. Shuai, *et al.*, "Thermal driven wavelength-selective optical switch based on magnetic polaritons coupling," J. Quant. Spectrosc. Radiat. Transf. **255**, 107230 (2020).
10. J. Qi, D. Zhang, Q. He, *et al.*, "Independent regulation of electrical properties of VO_2 for low threshold voltage electro-optic switch applications," Sens. Actuators A Phys. **335**, 113394 (2022).
11. Y. Guo, L. Zhu, S. Chen, *et al.*, "Dual-band polarized optical switch with opposite thermochromic properties to vanadium dioxide," Appl. Phys. Lett. **121**, 201102 (2022).
12. Z. Li, Z. Hu, J. Peng, *et al.*, "Ultrahigh infrared photoresponse from core-shell single-domain-VO_2/V_2O_5 heterostructure in nanobeam," Adv. Funct. Mater. **24**, 1821–1830 (2014).
13. Y. Jung, H. Han, A. Sharma, *et al.*, "Integrated hybrid VO_2–silicon optical memory," ACS Photonics **9**, 217–223 (2022).
14. Z. Ren, J. Xu, J. Liu, *et al.*, "Active and smart terahertz electro-optic modulator based on VO_2 structure," ACS Appl. Mater. Interfaces **14**, 26923–26930 (2022).
15. T. Kang, B. Fan, J. Qin, *et al.*, "Mid-infrared active metasurface based on Si/VO_2 hybrid meta-atoms," Photon. Res. **10**, 373–380 (2022).
16. M. Maaza, D. Hamidi, A. Simo, *et al.*, "Optical limiting in pulsed laser deposited VO_2 nanostructures," Opt. Commun. **285**, 1190–1193 (2012).
17. C. Wan, Z. Zhang, J. Salman, *et al.*, "Ultrathin broadband reflective optical limiter," Laser Photonics Rev. **15**, 2100001 (2021).
18. C. Wan, E. H. Horak, J. King, *et al.*, "Limiting optical diodes enabled by the phase transition of vanadium dioxide," ACS Photonics **5**, 2688–2692 (2018).
19. A. Tripathi, C. F. Ugwu, V. S. Asadchy, *et al.*, "Nanoscale optical non-reciprocity with nonlinear metasurfaces," Nat. Commun. **15**, 5077 (2024).
20. M. A. Kats, R. Blanchard, S. Zhang, *et al.*, "Vanadium dioxide as a natural disordered metamaterial: perfect thermal emission and large broadband negative differential thermal emittance," Phys. Rev. X **3**, 041004 (2014).
21. H. Guan, F. Ren, S. Liang, *et al.*, "Ultra-high transmission broadband tunable VO_2 optical limiter," Laser Photonics Rev. **17**, 2200653 (2023).
22. A. Howes, Z. Zhu, D. Curie, *et al.*, "Optical limiting based on Huygens' metasurfaces," Nano Lett. **20**, 4638–4644 (2020).
23. S. Im, H. Lee, and D. Kim, "Iterative opto-thermal analysis based on the temperature-dependent Drude–Lorentz dispersion model," in *Frontiers in Optics/Laser Science* (Optica Publishing Group, 2019).
24. M. Moskovits, I. Srnová-Šloufová, and B. Vlčková, "Bimetallic Ag-Au nanoparticles: extracting meaningful optical constants from the surface-plasmon extinction spectrum," J. Chem. Phys. **116**, 10435 (2002).
25. Y. Feng, Y. Wang, Z. Wang, *et al.*, "Damage mechanism and protective performance of vanadium dioxide under continuous-wave laser irradiation," Appl. Surf. Sci. **673**, 160847 (2024).
26. A. Tognazzi, M. Gandolfi, B. Li, *et al.*, "Opto-thermal dynamics of thin-film optical limiters based on the VO_2 phase transition," Opt. Mater. Express **13**, 41–52 (2022).
27. M. M. Qazilbash, M. Brehm, G. O. Andreev, *et al.*, "Infrared spectroscopy and nano-imaging of the insulator-to-metal transition in vanadium dioxide," Phys. Rev. B **79**, 075107 (2009).
28. C. Wan, Z. Zhang, D. Woolf, *et al.*, "On the optical properties of thin-film vanadium dioxide from the visible to the far infrared," Ann. Phys. **531**, 1900188 (2019).
29. C. Giannetti, "Dynamics of non-equilibrium states in solids induced by ultrashort coherent pulses," Ph.D. dissertation (Brescia University, 2004).